



Georgia Tech · Ivan Allen College of Liberal Arts

School of Modern Languages

Sentiment-linked quantification among adjectives

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We're setting up for a party and we need cups.

(1) The cups are *clean*.

~> All the cups are *clean*.



(2) The cups are *dirty*.

~> Some/most of the cups are *dirty*.



Variable quantification - why are *clean*/*dirty* different here?

“I was able to reach the safe, because...”

(1) The doors were open.

~> All the doors were open.



(2) The doors were open.

~> Some of the doors were open.



These inferences depend on the *context*, not just the specific word.

Preview of claims

- Sentences with definite plural subjects can be interpreted *universally* (“all”) or *existentially* (“some”) in different contexts.
- Semanticists offer different formalisms to capture such sentences – allowing key role for context/Question Under Discussion (QUD).
- **Q:** But why do *clean*/*dirty* favor different interpretations?
- **A:** Because *clean* is *good*, *dirty* is *bad* – with consequences for how the utterance addresses the QUD, via *All-Good*, *Any-Bad* Principle.

Sentiment

- Good/bad opposition is fundamental.
- Words carry emotional as well as denotational meaning.
 - *Clean* is good, *dirty* is bad (6.9/9 vs. 3.2/9 in Hedonometer).
 - *Adjectival* antonym pairs often organized around sentiment opposition.
 - Connects language to cognition; can be plotted on single axis.
- Discussed in psychology, NLP, corpus work – less so in semantics.
- **Claim:** Sentiment explains variable quantification.
 - Across individuals; degrees; dimensions.

Osgood et al 1957, Boucher and Osgood 1969, Sinclair 1991, Barrett 2006, Pang and Lee 2008, Dodds et al 2010, 2015; Glass 2023, Nouwen 2024, Gotzner and Mazzarella 2024...

The big picture

“The more closely we study actual linguistic forms, the more we [...] realize that they never express merely static, affectively neutral, concepts and judgments, but classes of concepts and judgments in which nuclear notions, capable of logical definition, are colored by unavowed dynamic and affective determinants.”

~ Edward Sapir (1944: 111-112)



- Quantification seems dispassionate, but it's shaped by emotion.
 - Connect logical, emotional dimensions of language.
- Illuminate words alongside human experience they describe.
- Converse with neighboring fields studying that experience.

Road map

- 👉 The All-Good, Any-Bad Principle distilled from social psychology
- Variable quantification
 - Across individuals
 - Across degrees
 - Across dimensions
- Discussion

The All-Good, Any-Bad Principle

- For something to be good, it must be all good.
- Any bad element makes it bad.
 - “A chain is as strong as its weakest link.”
 - Elaborate meal is ruined by a single cockroach.
 - Tragic flaw overshadows all a hero’s advantages.
- “All happy families are alike; each unhappy family is unhappy in its own way.”



Peeters and Czapinski 1990, Taylor 1991, Baumeister et al 2001, Rozin & Rozyman 2001

The All-Good, Any-Bad Principle

- **Bad** events elicit greater magnitude of emotion; also more causal reasoning, attribution of intention.
- “**Negativity dominance**” from social psychology.
- Flip side of the “**Pollyanna principle**”? (“look on the **bright** side”)
 - Word tokens in text are largely **positive** in sentiment.
- Homeostasis is mildly **positive**; thus any single **problem** seizes attention by **disrupting** it?

Boucher and Osgood 1969, Weiner 1985, Taylor 1991, Diamond 1997, Knobe 2003, ...

The All-Good, Any-Bad Principle

- Utterances address Question Under Discussion.
- “Is the current situation **okay**?” is a natural, common QUD.
 - Following the All-Good, Any-Bad Principle, this means:
 - “**Everything is okay**” or “**something is wrong.**”
 - (vs. “something is **okay**; everything is **wrong**”).
- I argue: This reasoning explains variable quantification.
 - Across individuals; degrees; and dimensions.

Road map

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S is “True-ish” if QUD is resolved the same as if S were true.

- “The doors were open.”
- (I could reach the safe.)

False	
False-ish	
True	

- “The doors were open.”
- (I could reach the safe.)

False	
True-ish	
True	

S is “True-ish” if QUD is resolved the same as if S were true.

- “The cups are **clean**.”
- (The situation is **okay**.)

True	
False-ish	
False	

- “The cups are **dirty**.”
- (The situation is **not okay**.)

False	
True-ish	
True	

The All-Good, Any-Bad Principle across individuals

- QUD: “Is the current situation **okay**?”
 - “The cups are **clean**” answers “Yes.”
 - For the situation to be **okay**, **all** the cups must be **clean**.
 - “The cups are **dirty**” answers “No.”
 - Even a **few dirty** cups make the situation **bad**.
- Predict: **Negative-sentiment** adjectives (**dirty**) are more likely to receive an existential (“some”) interpretations across individuals.

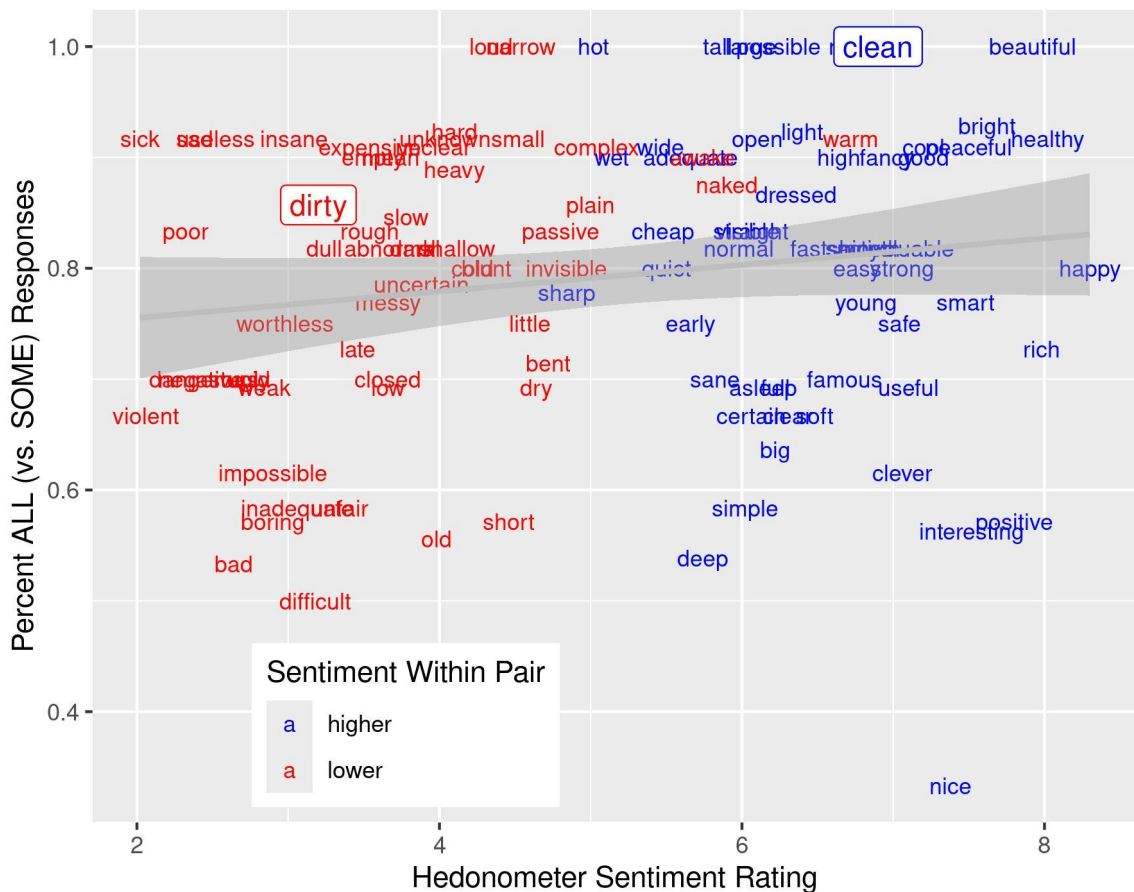
The All-Good, Any-Bad Principle across individuals

- *Experiment*: Your friend Ian is working with ten different cups. You ask Ian: “Are the cups {clean/dirty}?”
- Ian says, “Yes, the cups are {clean/dirty}.”
- What do you infer? [Binary judgment]
 - Some of them are {clean/dirty}.
 - All of them are {clean/dirty}.
- Predict: More “some” judgments for negative-sentiment adjectives.

The All-Good, Any-Bad Principle across individuals

- 40 native English speakers on Prolific; 55 antonym pairs (110 adjectives); 20% fillers with neutral/objective adjectives (*wooden*).
- Combined with artifact/human nouns that a person might be responsible for keeping in good condition.
 - clean/dirty cups
 - adequate/inadequate hotel rooms
 - clever/stupid advertisements
 - happy/sad customers

Percent ALL (vs. SOME) responses by Sentiment



As predicted: More “some” responses for **negative** adjectives.

Significant in mixed-effects logistic regression.

Subtle effect; “all” is strongly preferred overall.

The All-Good, Any-Bad Principle across individuals

(1) The cups are **clean**.



(2) The cups are **dirty**.



Q: Which adjectives tend to quantify universally vs. existentially?

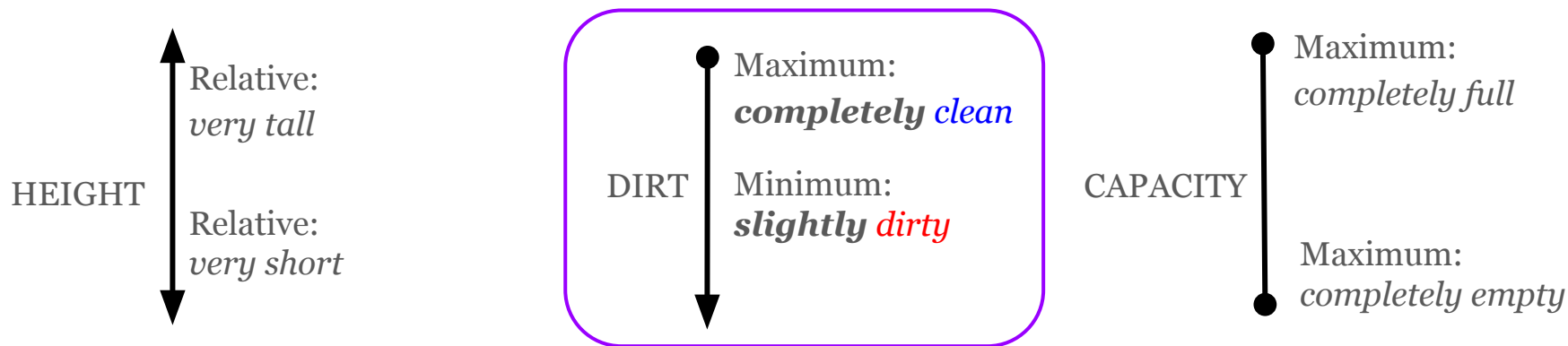
A: **Negative-sentiment** adjectives can be existential because *any bad* resolves QUD, “Is the current situation **okay**?” while *any good* doesn’t.

Road map

-  The All-Good, Any-Bad Principle distilled from social psychology
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The All-Good, Any-Bad Principle across degrees

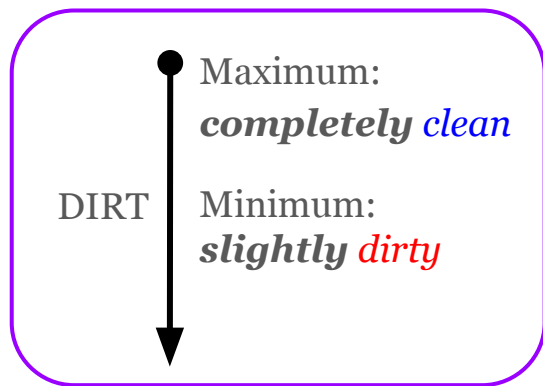
- Modifiers show: different adjectives evoke different types of scales.
- Max/min pairs show variable quantification across degrees.
 - Q: Which adjectives are associated with max vs. min standards?



Cruse 1980, Rotstein and Winter 2004, Kennedy 2007

The All-Good, Any-Bad Principle across degrees

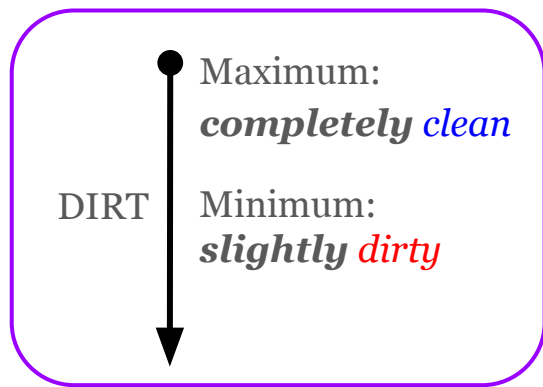
- The lexicon evolves to describe distinctions that people find useful.
- “Some **dirt**” (*any bad*) resolves QUD, “Is the current situation okay?” while “some **cleanness**” (*any good*) does not.



- More useful → more likely to be lexicalized.
- I propose: Minimum-standard adjectives (*dirty*) tend to be **negative** in sentiment.

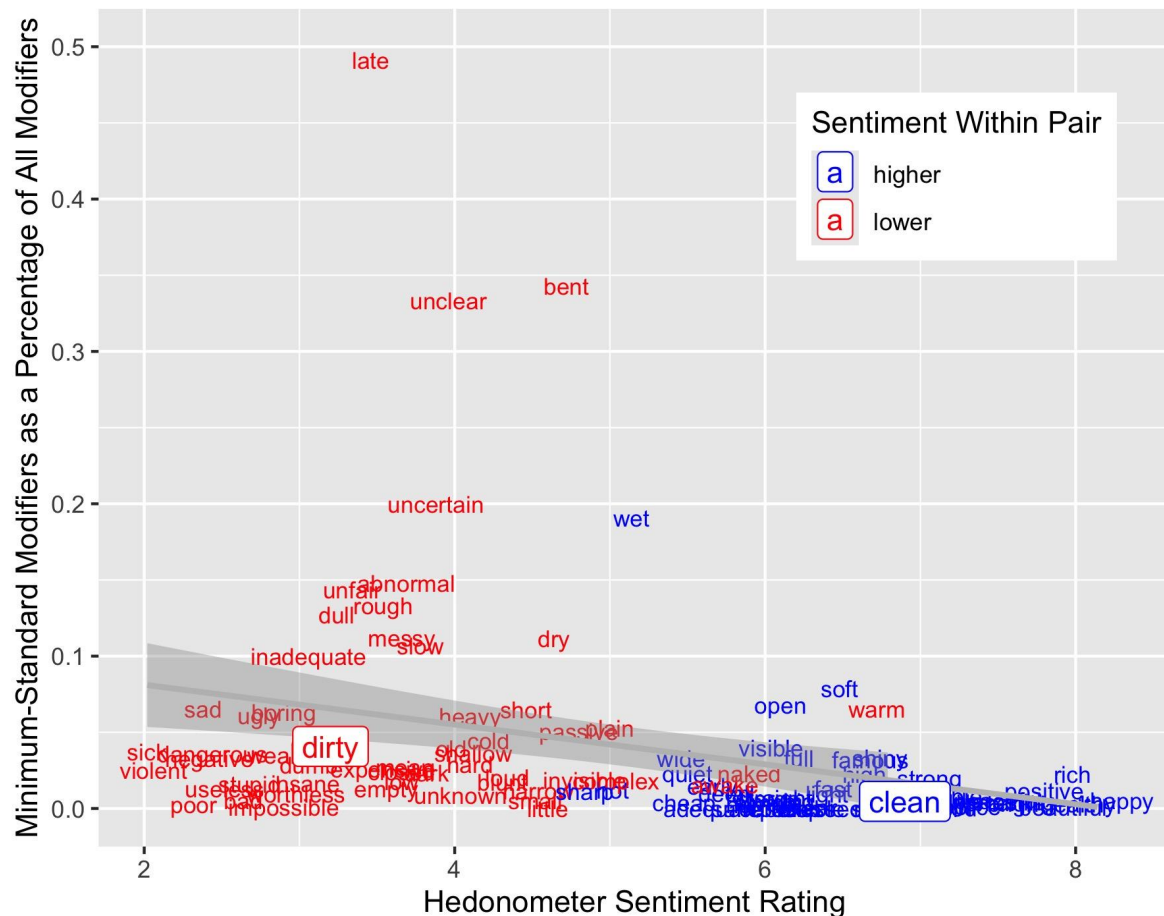
The All-Good, Any-Bad Principle across degrees

- For 110 adjectives, compute percent of the time modified by a min-standard modifier, as % of all modifiers.
- 100 million words from AskReddit.



- Min-standard modifiers: *(a) bit, (a) little, partly, partially, slightly*
- All other modifiers: *absolutely, almost, completely, extremely, very, so,*
- Predict: More min-standard modifiers (*slightly*) for negative adjectives (*dirty*).

Minimum-Standard Modifiers by Sentiment



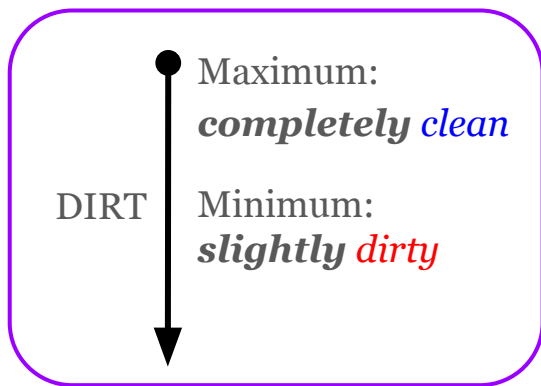
As predicted: **Negative** adjectives are more often modified by minimum-standard modifiers (*slightly*).

Significant in linear regression.

Consistent with claim that “**All-Good**, **Any-Bad**” Principle shapes lexicalization.

The All-Good, Any-Bad Principle across degrees

- **Q:** Which adjectives are associated with max vs. min standards?
- **A:** **Negative-sentiment** adjectives (*dirty*) are associated with minimum standards
 - via lexicalization pressure from All-Good, Any-Bad Principle.



Inspired by Cruse 1980

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The **All-Good**, **Any-Bad** Principle across dimensions

(1) My grandma is **healthy**.

~> **Healthy** across all dimensions.

✓	Good blood tests
✓	Good mobility

(2) My grandma is **sick**.

~> **Sick** along some dimension(s).

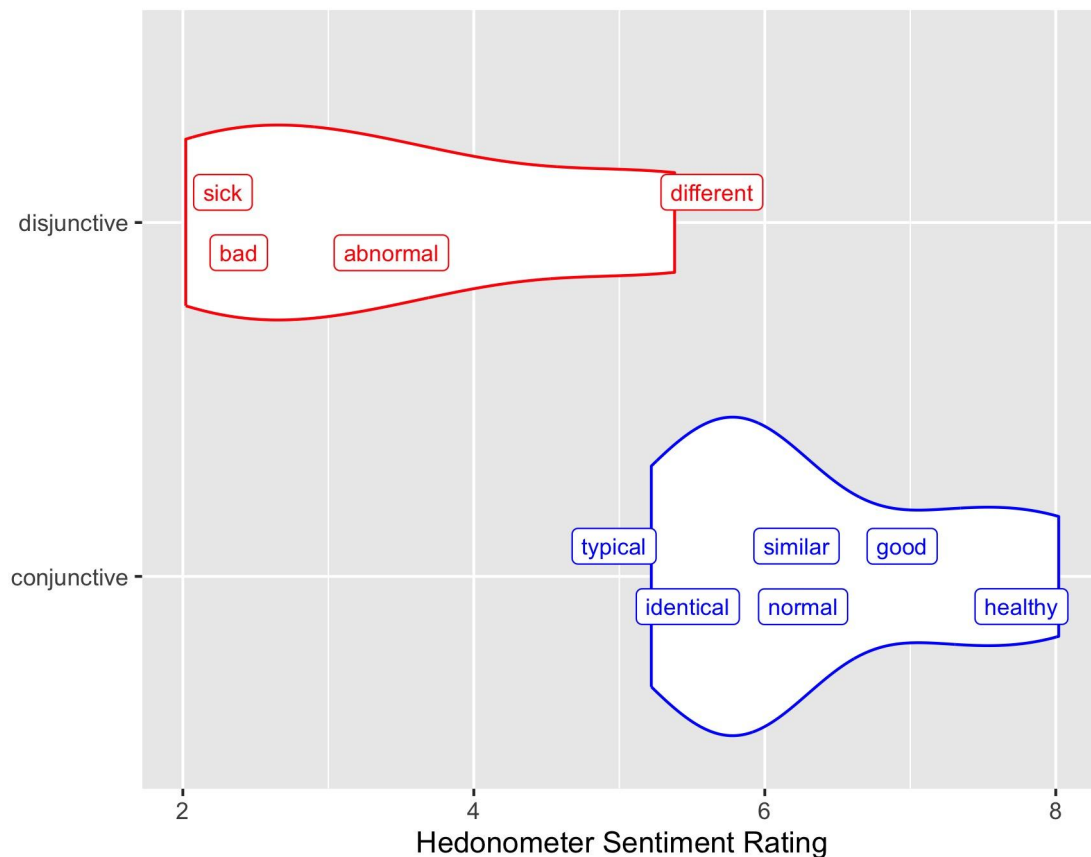
✗	Abnormal blood tests
✓	Good mobility

Q: Which adjectives are “conjunctive” (*all*) vs. “disjunctive” (*some*)?

The All-Good, Any-Bad Principle across dimensions

- “Some **sickness**” (*any bad*) resolves QUD, “*Is everything okay?*”; more useful → more likely to be lexicalized.
- “Some **health**” (*any good*) doesn’t resolve QUD; less useful → less likely to be lexicalized.
- Predict: Conjunctive multidimensional adjectives (*healthy*) should be **positive** in sentiment.
 - Their disjunctive antonyms (*sick*) should be **negative**.

Sentiment of Conjunctive vs. Disjunctive Multidimensional Adjectives from Sasson (2013)



10 adjectives classified by Sasson as conjunctive vs. disjunctive.

As predicted, conjunctive adjectives are more positive than disjunctive ones.

Consistent with claim that “**All-Good**, **Any-Bad**” Principle shapes lexicalization.

The All-Good, Any-Bad Principle across dimensions

(1) My grandma is **healthy**.

~> **Healthy** across all dimensions.

✓	Good blood tests
✓	Good mobility

(2) My grandma is **sick**.

~> **Sick** along some dimension(s).

✗	Abnormal blood tests
✓	Good mobility

Q: Which adjectives are “conjunctive” (*all*) vs. “disjunctive” (*some*)?

A: **Positive-sentiment adjectives** are conjunctive because they resolve QUD “*Is the current situation okay?*” with “Yes (it’s *all okay*).”

Road map

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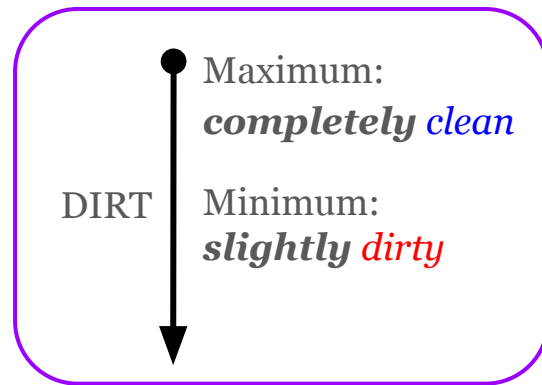
Summary

- **All-Good**, **Any-Bad** Principle from social psychology explains variable quantification across individuals; degrees; dimensions.



✓	Good blood tests
✓	Good mobility

✗	Abnormal blood tests
✓	Good mobility



What's new here

- The All-**Good**, Any-**Bad** Principle (Anna Karenina Principle) is longstanding.
- Cruse (1980) noted that max/min opposition among degrees is related to desirability (*perfectly **clean**, slightly **dirty***).
- This project is the first to observe effect of sentiment on quantification among individuals and dimensions as well as degrees.
- And the first to propose a unified explanation of these phenomena (tested empirically).

The big picture

- Quantification seems dispassionate, but it's shaped by emotion.
 - Quantities are often discussed as instantiating more/less than might be desired – no surprise that it's shaped by desirability.
 - Connect logical, emotional dimensions of language.
- Illuminate words alongside human experience they describe.
- Converse with neighboring fields studying that experience.

Thank you!

(with thanks to reviewers at *Natural Language Semantics* and, before that, *Journal of Semantics* for very helpful feedback)

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Relevance to this audience

- Quantification; truth; inferences drawn from words/sentences.
- Good/bad asymmetries and their inferential consequences.
- Text-based judgment experiments.
- Offering explanation for linguistic facts grounded in cognition rather than language itself.
- Autobiographical note: I got interested in this topic because I teach a gentle NLP course with a unit on sentiment.

E.g., Knobe 2003